Photovoltaic Performance Analysis of $RbGeI_3$ Absorber Based Perovskite Solar Cell using SCAPS-1D Simulation

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Abstract.

The outstanding rise of lead-halide perovskite solar cells has been tempered by the toxicity and environmental persistence of lead, motivating a sustained search for lead-free absorbers that retain favourable optoelectronic properties. Rubidium germanium iodide ($RbGeI_3$) is a promising candidate, offering a direct and tunable bandgap, strong optical absorption, and improved environmental compatibility. In this work, we report a comprehensive numerical optimization of an $RbGeI_3$-based perovskite solar cell using the one-dimensional Solar Cell Capacitance Simulator (SCAPS-1D, version 3.3.10). Eight planar heterojunction configurations were first screened by combining two electron transport layers (TiO_2 and PCBM) with four hole transport layers (NiO, CuI, CBTS, and Spiro-OMeTAD). The FTO/TiO_2/$RbGeI_3$/CuI/Au architecture delivered the best initial performance and was selected for further optimization. Eleven device parameters (absorber thickness, absorber defect density, dielectric constant, electron affinity, bandgap, conduction- and valence-band effective density of states, ETL and HTL thicknesses, and the interfacial defect densities at both heterojunctions) were varied sequentially under AM 1.5G illumination at 300 K. The optimized device reaches a power conversion efficiency (PCE) of 26.69 %, with an open-circuit voltage (V_{oc}) of 1.1127 V, a short-circuit current density (J_{sc}) of 30.32 mA/cm^2**, and a fill factor (FF) of 79.13 %.** The equilibrium energy band diagram confirms favourable band alignment at both heterojunctions. The results identify the TiO_2/CuI transport-layer pair as an efficient combination for environmentally benign $RbGeI_3$ photovoltaics and provide quantitative design guidelines for future experimental realization.

Keywords: lead-free perovskite; $RbGeI_3$; SCAPS-1D; numerical simulation; CuI hole transport layer; TiO_2 electron transport layer; defect density; interface engineering; solar cell optimization.

I. INTRODUCTION

Global electricity demand continues to climb as populations grow, economies industrialize, and urbanization accelerates. Fossil fuels still supply the bulk of this demand [1,2], but their depletion and the environmental cost of carbon emissions have made renewable energy a strategic priority rather than an alternative. Among renewable sources, solar energy stands out for its abundance and for the steady decline in the levelized cost of photovoltaic (PV) electricity, which has made solar the fastest-expanding renewable technology worldwide [3]. The global installed PV capacity surpassed 2.2 TW in 2024, and continued growth relies on identifying photovoltaic materials that combine high efficiency with low environmental impact and low fabrication cost.

Photovoltaic devices are commonly classified into three generations based on the absorber material and fabrication technology. First-generation crystalline silicon (c-Si) cells dominate the commercial market thanks to their high efficiency and long operational lifetime, but they require energy-intensive processing and rigid substrates. Second-generation thin-film technologies (including CdTe, CIGS, and amorphous silicon) reduce material consumption and offer flexible substrates, but their efficiencies remain modest and some rely on scarce or toxic elements. The third generation, which includes dye-sensitized solar cells, organic photovoltaics, quantum-dot cells, and perovskite solar cells (PSCs) [4,5], promises high efficiency at low processing cost and is therefore the focus of intense research activity.

Halide perovskites adopt an ABX_3 crystal structure in which the A-site hosts a monovalent cation, the B-site a divalent metal, and the X-site a halogen. Their attractiveness for photovoltaics stems from strong light absorption, continuously tunable bandgaps, ambipolar carrier transport with long diffusion lengths, and weak exciton binding [6,7]. Certified power conversion efficiencies (PCEs) for single-junction lead-based devices now exceed 26 % [8,9], and perovskite-on-silicon tandems have crossed 33 % [3]. The best absorbers, however, are methylammonium lead iodide ($MAPbI_3$) and formamidinium lead iodide ($FAPbI_3$), both of which contain lead. Lead raises contamination concerns at every stage of the device life cycle, from fabrication through operation to disposal [10,11], and the search for benign substitutes based on tin (Sn), germanium (Ge), bismuth (Bi), and antimony (Sb) has become a central research direction [12,13].

Tin-based perovskites initially attracted the most attention as lead substitutes because Sn^{2+} has a similar lone-pair electronic configuration to Pb^{2+} [14]. Unfortunately, Sn^{2+} readily oxidizes to Sn^{4+} in air, generating high densities of deep defects and accelerating device degradation [12,14]. Germanium-based perovskites have emerged as a complementary alternative because germanium offers a similar lone-pair electronic structure to lead, supports a direct and tunable bandgap, and is significantly less toxic [15]. Among Ge-based absorbers, rubidium germanium iodide ($RbGeI_3$) is particularly attractive thanks to its suitable bandgap (~1.3 eV), strong optical absorption coefficient, and good thermal stability [16,17,18]. Recent density-functional theory (DFT) calculations have confirmed that $RbGeI_3$ has a favourable direct bandgap and a high optical absorption coefficient comparable to that of $MAPbI_3$ [19,20,21], making it a promising candidate for single-junction photovoltaic applications.

Because experimental optimization of a multilayer photovoltaic device is both time-consuming and expensive, numerical simulation has become an indispensable tool for screening materials and parameters before fabrication. The Solar Cell Capacitance Simulator in one dimension (SCAPS-1D), developed at the Department of Electronics and Information Systems (ELIS) of Ghent University, Belgium, has been widely adopted for this purpose [22,23,24]. SCAPS-1D solves the Poisson equation together with the electron and hole continuity equations self-consistently across the device stack [22], capturing carrier generation, transport, and recombination under either dark or illuminated conditions. The software allows the user to vary material and interface parameters independently, which is essential for disentangling their individual contributions to device performance.

Although several SCAPS-1D studies of $RbGeI_3$ devices have been reported, most have optimized only a limited set of parameters or explored a single device configuration. The present work addresses this gap by combining an eight-configuration electron-transport-layer (ETL)/hole-transport-layer (HTL) screening with a sequential optimization of eleven absorber, interface, and transport-layer parameters for the most promising structure, FTO/TiO_2/$RbGeI_3$/CuI/Au. The optimized device is then compared with recently reported $RbGeI_3$-based and other lead-free Ge/Sn-based perovskite solar cells to evaluate its competitiveness and to identify remaining challenges for experimental realization. The remainder of this paper is organized as follows: Section II reviews the relevant literature and identifies the research gap; Section III describes the methodology; Section IV presents the results and discussion; Section V compares the proposed device with the state of the art; Section VI discusses limitations and future work; and Section VII concludes.

II. LITERATURE REVIEW AND RESEARCH GAP

Over the past five years, SCAPS-1D (often combined with first-principles DFT calculations) has become the workhorse numerical tool for designing and optimizing lead-free perovskite solar cells before experimental fabrication. Table I summarizes the most relevant recent studies on Ge-based and Sn-based absorbers, highlighting the device structure, the photovoltaic performance, and the principal limitation of each work.

TABLE I. Summary of recent SCAPS-1D studies on lead-free Ge-based and Sn-based perovskite solar cells.

Study (Year)	Device Structure	PCE (%)	Key Limitation
Pindolia et al. (2022) [25]	$FTO/TiO_2/RbGeI_3/NiO/Ag$	10.11	Single device; limited parametric sweep
Saikia et al. (2023) [26]	$FTO/TiO_2/CsGeI_3/CuI/Au$	10.80	No multi-HTL comparison
Bouazizi et al. (2022) [16]	FTO/PCBM/ $CsSn_{0.5}Ge_{0.5}I_3$/Spiro/Au	18.13	Mixed Ge/Sn absorber; thermal sensitivity
Loumachi et al. (2024) [28]	$ITO/C_6/RbGeI_3/CBTS/Au$	24.62	No interface defect optimization
Raj et al. (2025) [29]	$FTO/TiO_2/RbGeI_3$/ PEDOT:PSS/Au	18.44	Organic HTL limits stability
Pathak et al. (2026) [30]	$FTO/TiO_2/RbGeI_3$/Spiro-OMeTAD/Au	26.47	Ge^{2+} oxidation not addressed

Several observations emerge from this review. First, the experimental baseline for $RbGeI_3$ devices remains modest: Pindolia et al. reported a PCE of only 10.11 % for the $FTO/TiO_2/RbGeI_3/NiO/Ag$ architecture [25], and there is clearly substantial room for improvement through numerical optimization. Second, simulation-based studies have progressively pushed the predicted PCE of $RbGeI_3$ devices upward, from 10.11 % in 2022 to 24.62 % in 2024 [28] and 26.47 % in 2026 [30], but each of these works optimized only a limited subset of device parameters. Third, the ETL–HTL combinations explored for $RbGeI_3$ have been narrow: most studies used TiO_2 as the ETL and either Spiro-OMeTAD, NiO, or CBTS as the HTL, while the TiO_2/CuI combination adopted in this work has received comparatively little attention despite the well-documented advantages of CuI as an HTL.

Copper(I) iodide is a wide-gap ($E_g \approx 3.1$ eV), p-type transparent conductor whose valence-band maximum aligns closely with that of $RbGeI_3$, supporting efficient hole extraction [33,34]. Reported single-crystal hole mobilities reach order 10^2 $cm^2V^{-1}s^{-1}$, and unlike Spiro-OMeTAD or PEDOT:PSS the layer is solution-processable, inexpensive, and thermally robust [35,36]. Despite these advantages, the specific $TiO_2/RbGeI_3$/CuI stack, combined with a comprehensive optimization of the absorber and interface parameters, has not been reported in the open literature. The present study fills this gap by screening eight ETL–HTL combinations and then performing a sequential optimization of eleven parameters on the most promising architecture.

Beyond $RbGeI_3$, SCAPS-1D investigations of several related perovskite absorbers have established useful methodological precedents for transport-layer screening and defect-density optimization, including $CsPbI_3$ [37], the lead-free double perovskite Cs_2BiAgI_6 [38], and lead-free $CsSnI_3$ [39]. These studies demonstrate that the choice of ETL/HTL combination and the control of interfacial defect densities are at least as important as the absorber composition itself in determining the final PCE, which is the central premise of the present work.

III. METHODOLOGY

A. SCAPS-1D Simulation Tool

Numerical device simulation was performed with SCAPS-1D v3.3.10 [22,23], a one-dimensional solver originating from the ELIS group at Ghent University and distributed freely to academics. The code self-consistently integrates three coupled semiconductor equations across the device stack: the Poisson equation (1) for the electrostatic potential, the electron continuity equation (2) for generation, transport, and recombination of electrons, and the hole continuity equation (3) for the corresponding behaviour of holes.

$$\partial/\partial x \,[\, \varepsilon(x)\, \partial\psi/\partial x \,] = -\, q\, [\, p(x) - n(x) + N_D^{+}(x) - N_A^{-}(x) \pm \rho_t(x) \,] \qquad (1)$$

$$dJ_n/dx = G(x) - R(x), \qquad J_n = q\, \mu_n\, n\, E + q\, D_n\, dn/dx \qquad (2)$$

$$dJ_p/dx = R(x) - G(x), \qquad J_p = q\, \mu_p\, p\, E - q\, D_p\, dp/dx \qquad (3)$$

Here ε is the dielectric permittivity, ψ is the electrostatic potential, q is the elementary charge, n and p are the electron and hole concentrations, N_D and N_A are the ionized donor and acceptor densities, ρ_t is the trapped charge density, J_n and J_p are the electron and hole current densities, G is the carrier generation rate, R is the recombination rate, μ is the carrier mobility, and D is the diffusion coefficient. The user specifies the bandgap, electron affinity, dielectric constant, effective density of states, carrier mobilities, doping concentrations, and defect densities for each layer, together with the interface defect densities at each heterojunction. SCAPS-1D then returns the current density–voltage (J–V) characteristics, the quantum efficiency (QE) spectrum, and the equilibrium energy band diagram, from which the open-circuit voltage (V_{oc}), short-circuit current density (J_{sc}), fill factor (FF), and PCE are extracted. Fig. 1 shows the graphical user interface of SCAPS-1D version 3.3.10 as used in this study.

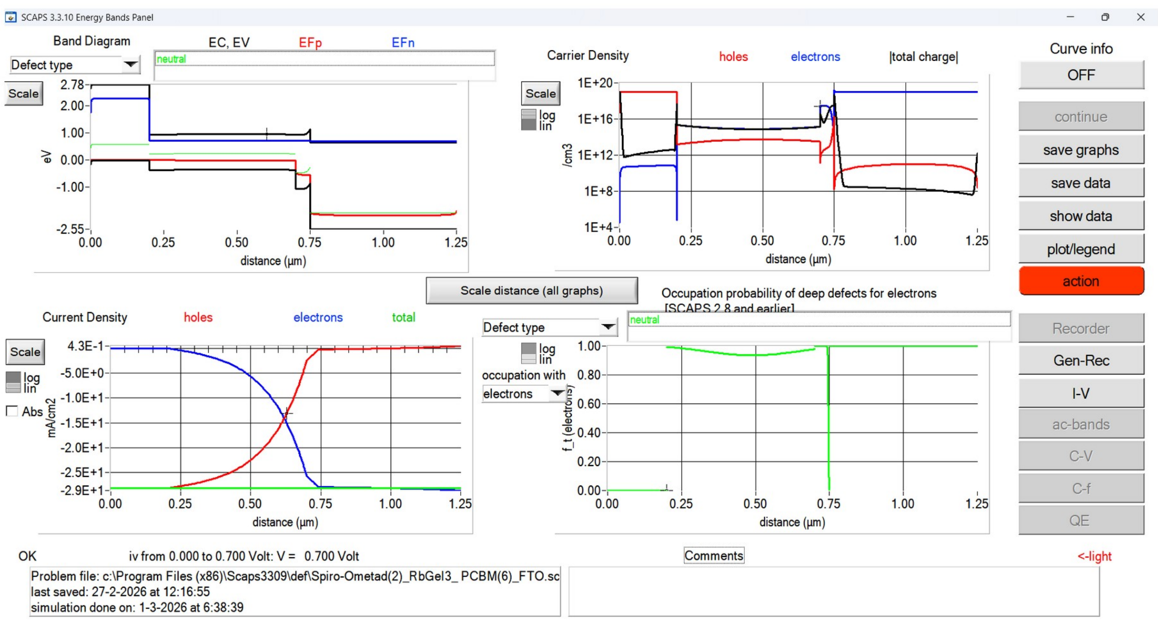

FIG. 1. Graphical user interface of SCAPS-1D (version 3.3.10) used for device simulation, showing the energy band diagram, carrier density, current density, and defect occupation panels at a bias of 0.700 V.

B. Device Structure and Working Principle

The proposed device follows a conventional n-i-p planar heterojunction architecture: FTO/TiO_2/$RbGeI_3$/CuI/Au. Fluorine-doped tin oxide (FTO) acts as the transparent front electrode [40], titanium dioxide (TiO_2) serves as the ETL, $RbGeI_3$ is the photoactive absorber, copper iodide (CuI) acts as the HTL, and gold (Au) provides the rear ohmic contact. This layer sequence was chosen because it provides efficient charge separation, rapid carrier transport, and suitable energy-level alignment while eliminating the environmental and health concerns associated with lead-based perovskite absorbers.

When sunlight is incident on the front surface of the device, photons first pass through the transparent FTO electrode and the thin TiO_2 layer before reaching the $RbGeI_3$ absorber. Photons with energies greater than the absorber bandgap are absorbed within the $RbGeI_3$ layer, generating electron–hole pairs. The built-in electric field established across the heterojunction then separates these photogenerated carriers: electrons are transported from the absorber into the TiO_2 electron transport layer and subsequently collected by the FTO electrode, while holes move toward the CuI hole transport layer before being collected by the Au back contact. This directional carrier transport minimizes carrier recombination and contributes to improved photovoltaic performance.

TiO_2 was chosen as the ETL because of its wide bandgap ($E_g \approx 3.2$ eV), suitable conduction-band position, chemical stability, and well-documented compatibility with perovskite absorbers [41,42]. CuI was selected as the HTL because of its high hole mobility, wide bandgap ($E_g \approx 3.1$ eV), low fabrication cost, and favourable valence-band alignment with $RbGeI_3$ [33,34]. Gold was used as the rear metal contact because of its high work function (~5.1 eV), excellent electrical conductivity, and superior chemical stability, which together establish an ohmic contact with the CuI hole transport layer and reduce contact resistance [43].

C. Material Parameters and Simulation Conditions

The accuracy of any SCAPS-1D simulation depends critically on the selection of appropriate material parameters for each layer of the solar cell. The initial electrical and optical parameters used in this study were collected from previously published experimental and numerical studies of $RbGeI_3$ and related lead-free perovskites [16,17,25,26]. These values were then used as the starting point for the sequential optimization. Table II summarizes the initial input parameters for each of the four semiconductor layers, and Table III lists the interface defect densities applied at the two heterojunctions.

TABLE II. Initial electrical and optical parameters used in the SCAPS-1D simulation of the proposed FTO/TiO_2/$RbGeI_3$/CuI/Au solar cell.

Parameter	FTO	TiO_2	$RbGeI_3$	CuI
Thickness (nm)	500	30	400	100
Bandgap, E_g (eV)	3.2	3.2	1.31	3.1
Electron affinity, χ (eV)	4.4	4.0	3.9	2.1
Dielectric permittivity, ε_r	9	9	23.01	6.5
CB effective DOS, N_C (cm^{-3})	2.2×10^{18}	2×10^{18}	1.4×10^{19}	2.8×10^{19}
VB effective DOS, N_V (cm^{-3})	1.8×10^{19}	1.8×10^{19}	2.8×10^{19}	1×10^{19}

Electron mobility, μ_e (cm^2/Vs)	20	20	28.6	100
Hole mobility, μ_p (cm^2/Vs)	10	10	27.3	43.9
Donor density, N_D (cm^{-3})	1×10^{19}	9×10^{16}	1×10^{9}	0
Acceptor density, N_A (cm^{-3})	0	0	1×10^{9}	1×10^{18}
Defect density, N_t (cm^{-3})	1×10^{14}	1×10^{15}	1×10^{15}	1×10^{15}

TABLE III. Initial interface defect density applied at each heterojunction.

Interface	**N_t (cm^{-2})**
$TiO_2/RbGeI_3$	1×10^{14}
$RbGeI_3$/CuI	1×10^{14}

All simulations were performed under the standard AM 1.5G solar spectrum at an incident power density of 100 mW/cm² (1000 W/m²) and a temperature of 300 K, following standard test conditions (STC). During each optimization step, only one parameter was varied while all others were held constant, and the optimum value obtained from each step was carried forward as the input for the next step. This sequential one-at-a-time strategy is widely used in SCAPS-based studies because it allows the contribution of each parameter to be evaluated cleanly, without confounding effects from simultaneous variations.

D. Optimization Procedure

Fig. 2 illustrates the sequential optimization procedure adopted in this study. The process begins with the initial unoptimized FTO/TiO_2/$RbGeI_3$/CuI/Au device (configuration D4 selected from the screening step described in Section IV A). Each of the eleven parameters listed in Table XVI is then swept in turn, with the optimum value from each step carried forward as the input to the next. After all eleven parameters have been optimized, the final J–V curve, QE spectrum, and equilibrium energy band diagram are extracted from the optimized device for analysis. The overall research methodology is summarized in Fig. 2.

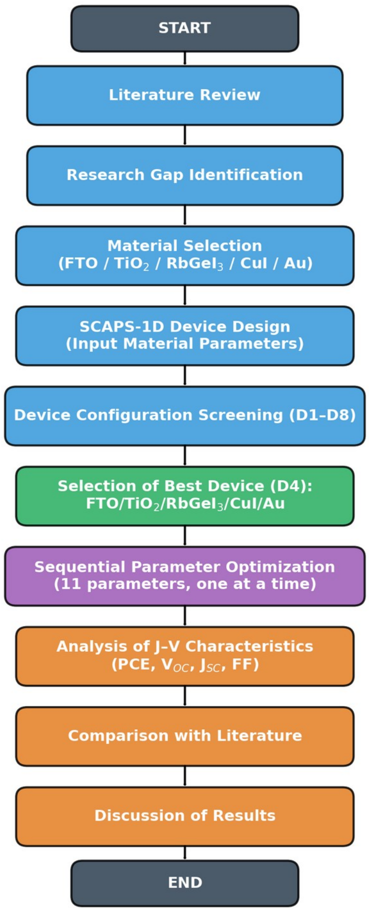

FIG. 2. Overall research methodology adopted in this study, from the initial literature review through material selection, device screening, parametric optimization, and final performance comparison.

E. Performance Metrics

Four metrics extracted from the simulated J–V curve were used to evaluate device performance. The open-circuit voltage V_{oc} is the maximum voltage generated when no external current flows and reflects the splitting of the electron and hole quasi-Fermi levels. The short-circuit current density J_{sc} is the current density at zero applied voltage and is governed by the absorption and collection of photogenerated carriers. The fill factor FF measures the rectangularity of the J–V curve and is given by

$$FF = (V_{mp} \times J_{mp}) / (V_{oc} \times J_{sc}) \quad (4)$$

where V_{mp} and J_{mp} are the voltage and current density at the maximum power point. The power conversion efficiency PCE represents the fraction of incident solar power converted into electrical power and is defined as

$$PCE = (V_{oc} \times J_{sc} \times FF) / P_{in} \times 100\ \% \quad (5)$$

where P_{in} is the incident optical power density (100 mW/cm² under AM 1.5G). These four parameters were tracked throughout the optimization process and used to compare the proposed device with previously reported $RbGeI_3$ and related lead-free perovskite solar cells.

IV. RESULTS AND DISCUSSION

A. Screening of Device Configurations

Eight planar heterojunction configurations were constructed by combining the two ETL candidates (TiO_2 and PCBM) with four HTL candidates (NiO, CuI, CBTS, and Spiro-OMeTAD), all sharing the same $RbGeI_3$ absorber. The eight devices (labelled D1 through D8) were simulated under identical conditions, and the resulting photovoltaic parameters are summarized in Table IV. Device D4, corresponding to the FTO/TiO_2/$RbGeI_3$/CuI/Au architecture, achieved the highest initial PCE of 19.79 %, with V_{oc} = 0.812 V, J_{sc} = 30.46 mA/cm^2, and FF = 79.98 %. D4 was therefore selected as the baseline for all subsequent optimization.

The advantage of D4 over the other configurations can be traced to two factors. First, the favourable conduction-band alignment between TiO_2 (χ = 4.0 eV) and $RbGeI_3$ (χ = 3.9 eV) produces only a small conduction-band offset (~0.10 eV) [41,44], which supports efficient electron extraction without introducing a significant barrier. Second, the small valence-band offset between $RbGeI_3$ and CuI ($\Delta E_v \approx 0.1$ eV) allows holes to be extracted with minimal resistance, while the large conduction-band offset at the same interface ($\Delta E_c \approx 1.8$ eV) acts as an effective electron-blocking barrier [34]. Together, these two features suppress interfacial recombination and support the high FF and V_{oc} observed for D4.

TABLE IV. Initial photovoltaic performance of the eight screened device configurations. ★ Selected as the baseline for full parametric optimization.

Dev.	Architecture	V_{oc} (V)	J_{sc} (mA/cm^2)	FF (%)	PCE (%)
D1	FTO/PCBM/ $RbGeI_3$/NiO/Au	0.81	30.10	83.68	20.52
D2	FTO/TiO_2/ $RbGeI_3$/NiO/Au	0.82	30.46	84.48	21.08
D3	FTO/PCBM/ $RbGeI_3$/CuI/Au	0.81	30.10	83.96	20.55
D4 ★	FTO/TiO_2/ $RbGeI_3$/CuI/Au	0.812	30.46	79.98	19.79
D5	FTO/PCBM/ $RbGeI_3$/ CBTS/Au	0.82	30.20	82.36	20.30
D6	FTO/TiO_2/ $RbGeI_3$/ CBTS/Au	0.82	30.55	82.13	20.68
D7	FTO/PCBM/ $RbGeI_3$/Spiro-OMeTAD/Au	0.81	30.07	85.01	20.77
D8	FTO/TiO_2/ $RbGeI_3$/Spiro-OMeTAD/Au	0.82	30.46	75.76	18.92

B. Absorber Thickness

The absorber thickness governs the trade-off between optical absorption and carrier collection. A thicker layer absorbs more photons but also forces photogenerated carriers to travel farther, which raises the probability of bulk recombination. To map this trade-off, the $RbGeI_3$ thickness was swept from 0.3 µm to 1.0 µm while the other parameters were held constant. The results are shown in Fig. 3 and Table V. The PCE rose sharply from 18.57 % at 0.3 µm to a peak of 20.70 % at 0.7 µm and then declined gently to 20.46 % at 1.0 µm, in line with the expected onset of recombination losses in thicker absorbers. J_{sc} increased monotonically from 28.14 to 34.11 mA/cm^2 because more photons were absorbed, while V_{oc} and FF decreased gradually as the absorber became thicker (V_{oc} fell from 0.823 V to 0.767 V, and FF from 80.18 % to 78.20 %). A thickness of 700 nm was therefore adopted as the optimum, since it maximizes efficiency without paying the cost of an unnecessarily thick absorber.

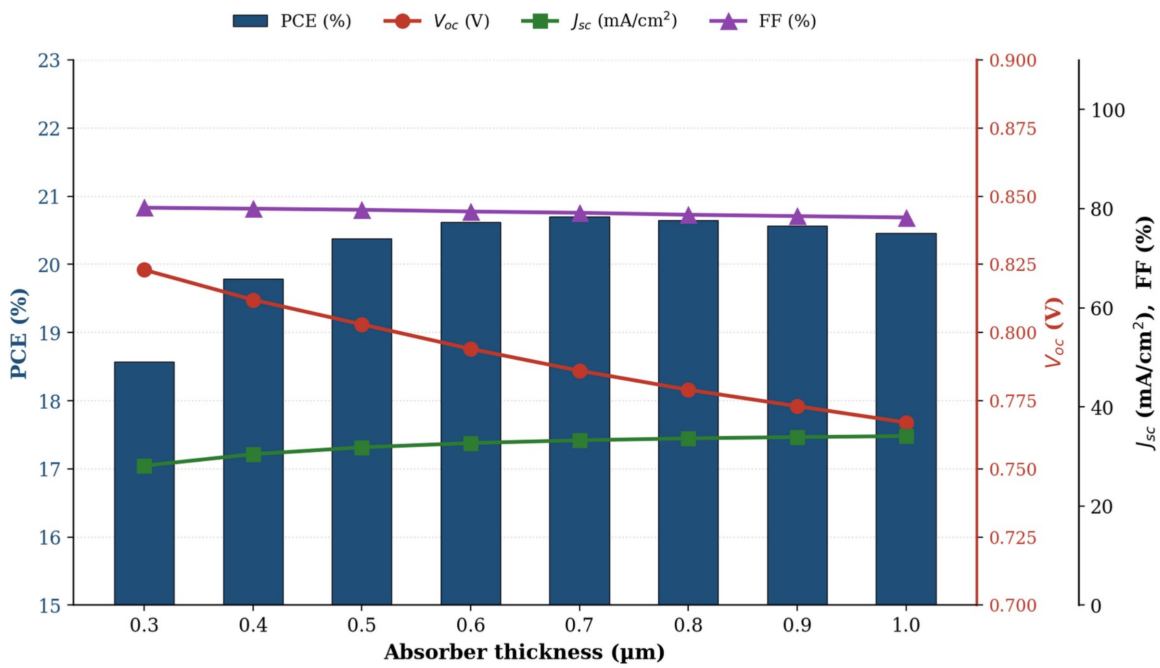

FIG. 3. Variation of PCE, V_{oc}, J_{sc}, and FF with $RbGeI_3$ absorber thickness. The PCE peaks at 0.7 µm, beyond which recombination losses offset the gain in optical absorption.

TABLE V. Variation of PCE, V_{oc}, J_{sc}, and FF with $RbGeI_3$ absorber thickness.

Thickness (µm)	PCE (%)	V_{oc} (V)	J_{sc} (mA/cm²)	FF (%)
0.3	18.57	0.823	28.14	80.18
0.4	19.79	0.812	30.46	79.98
0.5	20.38	0.803	31.84	79.76
0.6	20.62	0.794	32.70	79.41
0.7 ★	20.70	0.786	33.27	79.15
0.8	20.65	0.779	33.65	78.75
0.9	20.57	0.773	33.91	78.48
1.0	20.46	0.767	34.11	78.20

C. Absorber Defect Density

Bulk defect density is one of the most damaging parameters in perovskite devices because trap states act as Shockley–Read–Hall (SRH) recombination centres that capture and annihilate photogenerated carriers. To quantify this effect, the defect density of the $RbGeI_3$ layer was varied from 10^{12} to 10^{18} cm^{-3} while all other parameters were held constant. The results are shown in Fig. 4 and Table VI. Below 10^{15} cm^{-3} the device performance was flat, with PCE remaining close to 20.5 %. Above 10^{15} cm^{-3} the degradation became severe:

PCE dropped to 13.78 % at 10^{17} cm^{-3} and to only 7.58 % at 10^{18} cm^{-3}, while V_{oc} fell from 0.86 V to 0.55 V. J_{sc} also decreased substantially at high defect densities (from 30.46 to 26.61 mA/cm^2) because the trapped carriers recombined before being collected. Although the best simulated result was obtained at 10^{12} cm^{-3}, such a low defect density is not realistic for an experimental Ge-based perovskite film, where Ge^{2+} oxidation typically introduces defect densities of at least 10^{13}–10^{14} cm^{-3} [30,31]. A defect density of 10^{14} cm^{-3} was therefore selected as a practical compromise that retains more than 99 % of the maximum efficiency.

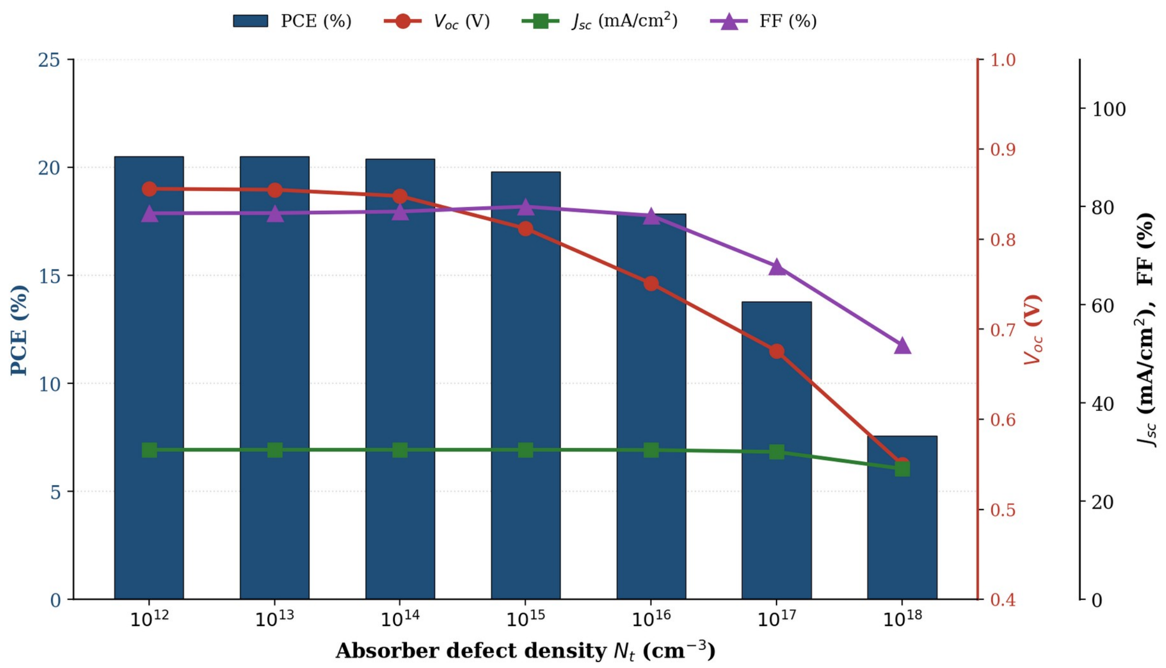

FIG. 4. Variation of PCE, V_{oc}, J_{sc}, and FF with $RbGeI_3$ absorber defect density. The device is insensitive to defect densities below 10^{15} cm^{-3} but degrades sharply at higher values due to SRH recombination.

TABLE VI. Variation of PCE, V_{oc}, J_{sc}, and FF with $RbGeI_3$ absorber defect density.

N_t (cm^{-3})	PCE (%)	V_{oc} (V)	J_{sc} (mA/cm^2)	FF (%)
10^{12}	20.51	0.856	30.46	78.62
10^{13}	20.50	0.855	30.46	78.66
10^{14} ★	20.40	0.848	30.46	78.97
10^{15}	19.79	0.812	30.46	79.98
10^{16}	17.86	0.751	30.42	78.15
10^{17}	13.78	0.676	30.03	67.87
10^{18}	7.58	0.550	26.61	51.79

D. Dielectric Constant

The dielectric constant of the absorber influences the electrostatic potential distribution, the screening of charged defects, and the electric field across the depletion region. To evaluate its effect, the relative permittivity ε_r of the $RbGeI_3$ layer was varied from 15 to 23.1 while all other parameters were held constant. The results are shown in Fig. 5 and Table VII. Increasing ε_r caused only a marginal reduction in device performance: PCE decreased from 19.92 % at $\varepsilon_r = 15$ to 19.79 % at $\varepsilon_r = 23.1$, while V_{oc} dropped slightly from 0.819 V to 0.812 V. J_{sc} remained almost unchanged at 30.46 mA/cm^2, and FF showed a very small improvement from 79.87 % to 79.98 %. The decrease in PCE can therefore be attributed almost entirely to the small reduction in V_{oc}, which arises because a higher dielectric constant reduces the built-in electric field across the depletion region. Although the effect is small within the studied range, the lower dielectric constant is slightly more favourable, and a value of $\varepsilon_r = 15$ was adopted for the optimized device.

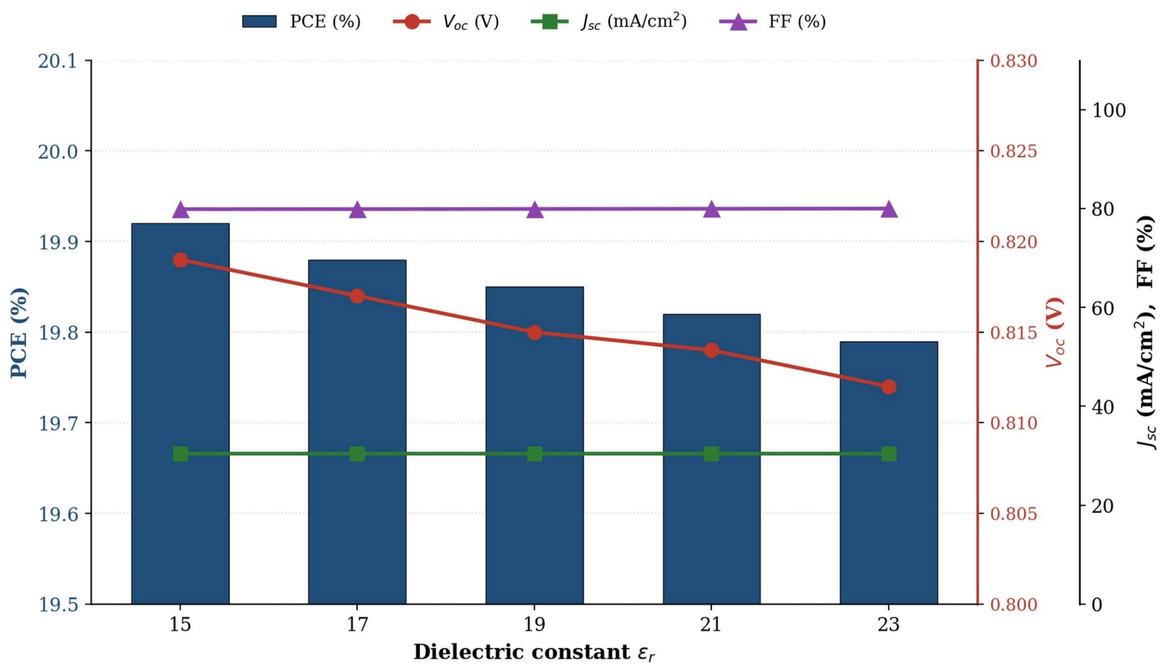

FIG. 5. Variation of PCE, V_{oc}, J_{sc}, and FF with the $RbGeI_3$ absorber dielectric constant. The effect is small within the studied range, with lower ε_r values being slightly more favourable.

TABLE VII. Variation of PCE, V_{oc}, J_{sc}, and FF with $RbGeI_3$ absorber dielectric constant.

ε_r	PCE (%)	V_{oc} (V)	J_{sc} (mA/cm²)	FF (%)
15 ★	19.92	0.819	30.46	79.87
17	19.88	0.817	30.46	79.88
19	19.85	0.815	30.46	79.91
21	19.82	0.814	30.46	79.94
23	19.79	0.812	30.46	79.98

E. Electron Affinity

The electron affinity χ of the absorber directly controls the conduction-band offset at the absorber/ETL interface and is therefore one of the most sensitive parameters in the device. To evaluate its effect, χ was swept from 3.9 eV to 4.2 eV while all other parameters were held constant. The results are shown in Fig. 6 and Table VIII. V_{oc} dropped sharply from 0.812 V at χ = 3.9 eV to 0.710 V at χ = 4.2 eV, while J_{sc} remained nearly constant at 30.46 mA/cm^2. The PCE decreased from 19.79 % at 3.9 eV to 17.81 % at 4.2 eV, with the loss driven almost entirely by the voltage. FF actually increased slightly from 79.98 % to 82.33 % over the same range, but this was insufficient to offset the V_{oc} loss. The optimum value of χ = 3.9 eV gives a small positive conduction-band spike ($\Delta E_c \approx 0.10$ eV) at the TiO_2/$RbGeI_3$ interface, which is known to suppress electron back-transfer without significantly impeding extraction. Higher values of χ move the absorber conduction band below that of TiO_2, creating a cliff that facilitates recombination and reduces V_{oc}.

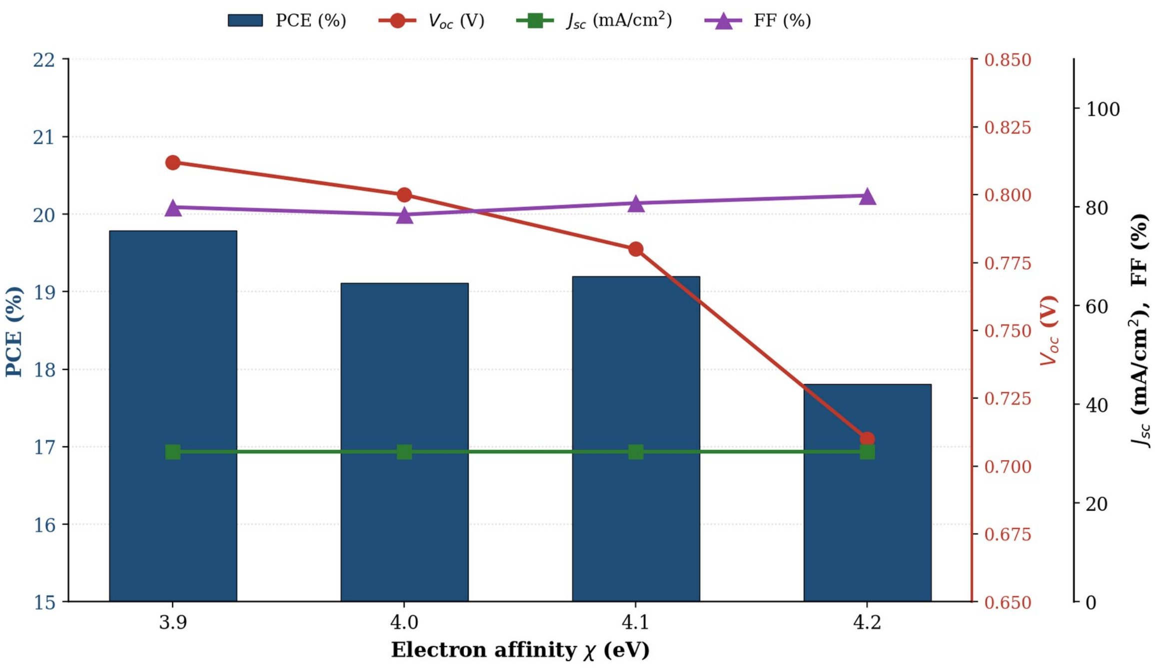

FIG. 6. Variation of PCE, V_{oc}, J_{sc}, and FF with the $RbGeI_3$ absorber electron affinity. V_{oc} drops sharply as χ increases beyond 3.9 eV due to an unfavourable conduction-band cliff at the $TiO_2/RbGeI_3$ interface.

TABLE VIII. Variation of PCE, V_{oc}, J_{sc}, and FF with $RbGeI_3$ absorber electron affinity.

χ (eV)	PCE (%)	V_{oc} (V)	J_{sc} (mA/cm²)	FF (%)
3.9 ★	19.79	0.812	30.46	79.98
4.0	19.11	0.800	30.46	78.47
4.1	19.20	0.780	30.46	80.79
4.2	17.81	0.710	30.46	82.33

F. Interfacial Defect Densities

Interfacial defects are particularly harmful because they sit directly in the carrier extraction path. Two interfaces were examined: $TiO_2/RbGeI_3$ (absorber/ETL) and $RbGeI_3/CuI$ (absorber/HTL). At the absorber/ETL interface (Fig. 7, Table IX), increasing the defect density from 10^{12} to 10^{18} cm^{-2} caused a steep collapse of performance: PCE fell from 21.04 % to 8.40 %, V_{oc} from 0.827 V to 0.439 V, J_{sc} from 30.46 to 28.09 mA/cm^2, and FF from 83.52 % to 68.11 %. This degradation occurs because interfacial defects act as recombination centres where photogenerated electrons and holes recombine before being collected at the electrodes.

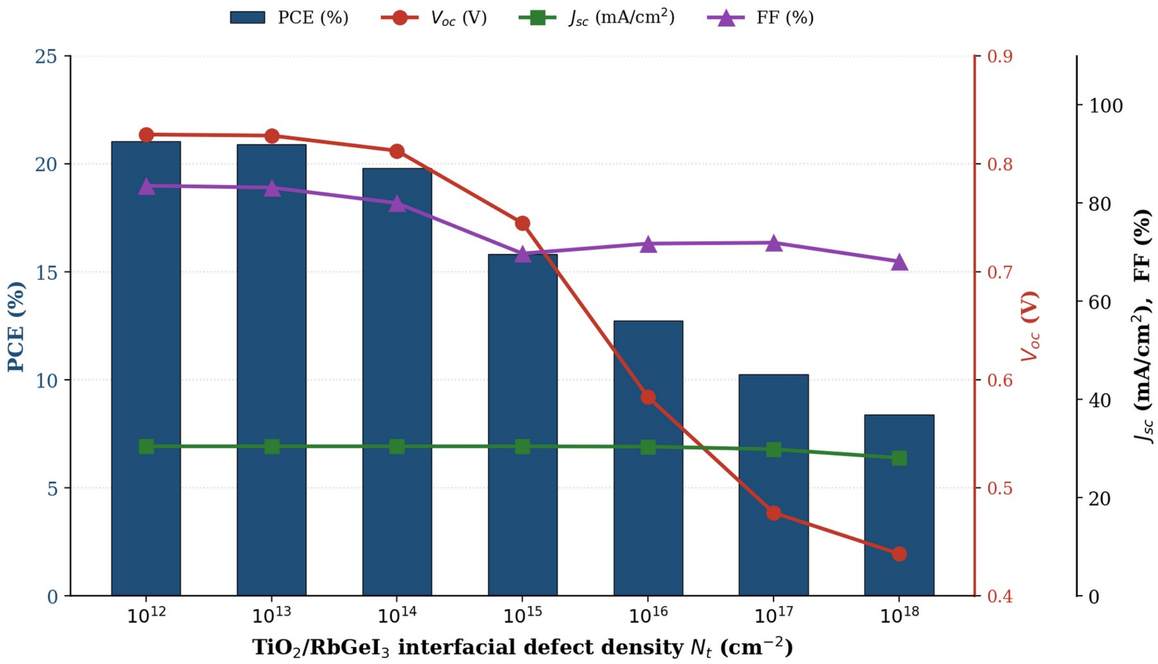

FIG. 7. Variation of PCE, V_{oc}, J_{sc}, and FF with the $TiO_2/RbGeI_3$ interfacial defect density. Performance degrades sharply above 10^{14} cm^{-2}, indicating high sensitivity to absorber/ETL interface quality.

TABLE IX. Variation of PCE, V_{oc}, J_{sc}, and FF with $TiO_2/RbGeI_3$ interfacial defect density.

N_t (cm^{-2})	PCE (%)	V_{oc} (V)	J_{sc} (mA/cm^2)	FF (%)
10^{12}	21.04	0.827	30.46	83.52
10^{13}	20.91	0.826	30.46	83.14
10^{14} ★	19.79	0.812	30.46	79.98
10^{15}	15.81	0.745	30.45	69.70
10^{16}	12.74	0.584	30.39	71.73
10^{17}	10.24	0.477	29.86	71.91
10^{18}	8.40	0.439	28.09	68.11

The HTL/absorber interface (Fig. 8, Table X) proved far more tolerant than the absorber/ETL interface: the device output stayed flat up to 10^{15} cm^{-2} (PCE = 19.78 %, V_{oc} = 0.811 V) and only degraded noticeably at higher defect densities, reaching 18.19 % at 10^{18} cm^{-2}. This asymmetry is consistent with the band alignment at the two junctions. The large conduction-band offset at the $CuI/RbGeI_3$ interface ($\Delta E_c \approx 1.8$ eV) acts as an electron-blocking barrier that limits the impact of interfacial traps, whereas the small offset at the $TiO_2/RbGeI_3$ interface ($\Delta E_c \approx 0.10$ eV) offers less protection. A realistic interface defect density of 10^{14} cm^{-2} was adopted at this junction as well, matching the value used at the absorber/ETL interface.

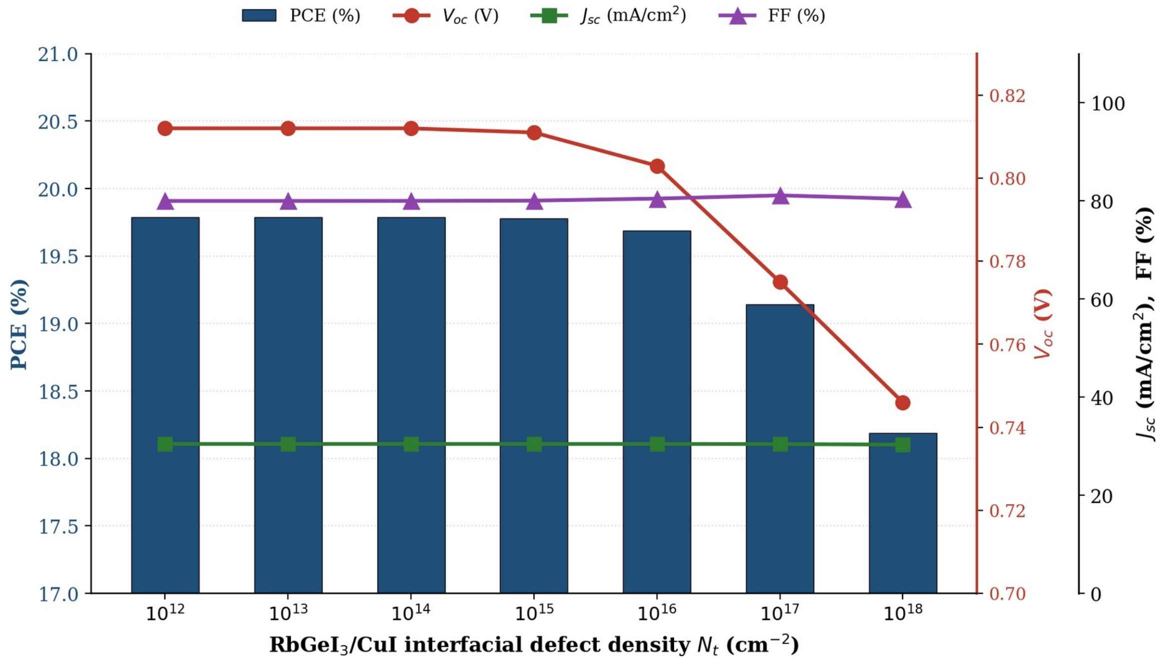

FIG. 8. Variation of PCE, V_{oc}, J_{sc}, and FF with the $RbGeI_3$/CuI interfacial defect density. The HTL/absorber interface is more tolerant than the absorber/ETL interface, thanks to the large conduction-band offset at the CuI/$RbGeI_3$ junction.

TABLE X. Variation of PCE, V_{oc}, J_{sc}, and FF with $RbGeI_3$/CuI interfacial defect density.

N_t (cm^{-2})	PCE (%)	V_{oc} (V)	J_{sc} (mA/cm^2)	FF (%)
10^{12}	19.79	0.812	30.46	79.97
10^{13}	19.79	0.812	30.46	79.97
10^{14} ★	19.79	0.812	30.46	79.98
10^{15}	19.78	0.811	30.46	80.04
10^{16}	19.69	0.803	30.46	80.45
10^{17}	19.14	0.775	30.43	81.11
10^{18}	18.19	0.746	30.33	80.39

G. Absorber Bandgap

The bandgap of the absorber determines the spectral range of photon absorption and therefore sets a fundamental trade-off between V_{oc} and J_{sc}: a wider bandgap increases V_{oc} but excludes an increasing fraction of the solar spectrum, reducing J_{sc}. To map this trade-off, the $RbGeI_3$ bandgap was swept from 1.3 eV to 1.7 eV while all other parameters were held constant. The results are shown in Fig. 9 and Table XI. V_{oc} rose steadily from 0.803 V at 1.3 eV to 0.997 V at 1.7 eV, as expected from the increased quasi-Fermi-level splitting. J_{sc}, however, dropped sharply from 30.79 mA/cm^2 at 1.3 eV to 17.95 mA/cm^2 at 1.7 eV because the wider gap excludes an increasing fraction of the solar spectrum. The PCE peaked at 19.87 % for a bandgap of 1.4 eV, which represents the best compromise between voltage and current. This value is also close to the optimum bandgap of a single-junction cell predicted by the Shockley–Queisser detailed-balance limit (~1.34 eV) [45], and was therefore adopted for the final device.

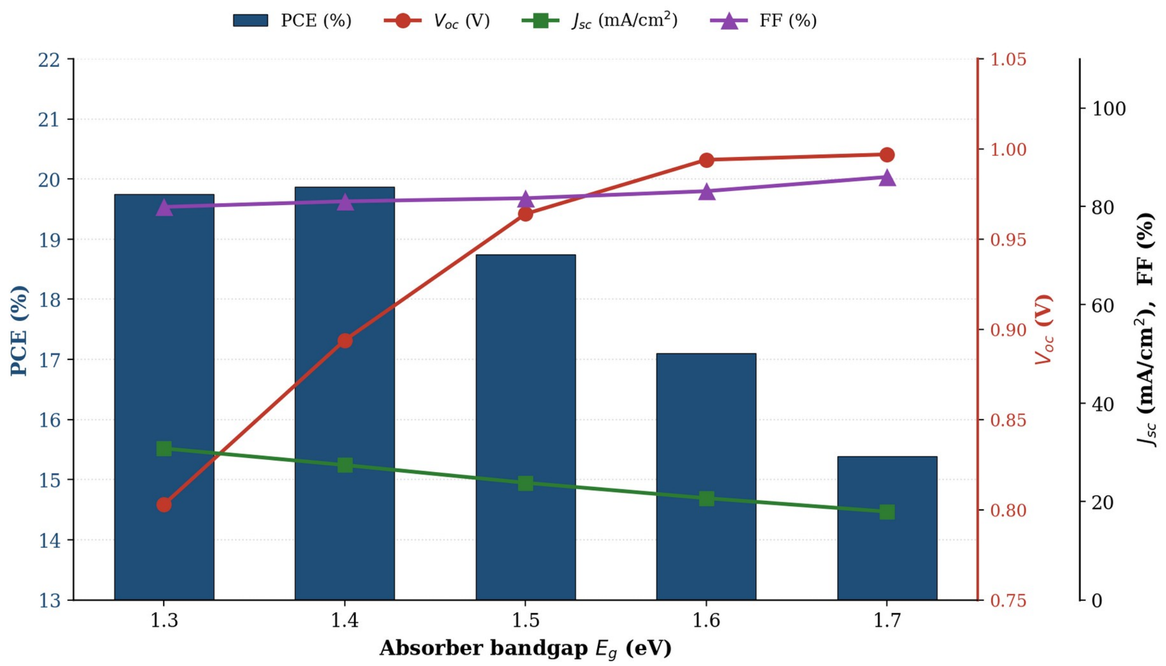

FIG. 9. Variation of PCE, V_{oc}, J_{sc}, and FF with the $RbGeI_3$ absorber bandgap. The PCE peaks at 1.4 eV, close to the Shockley–Queisser optimum for a single-junction cell.

TABLE XI. Variation of PCE, V_{oc}, J_{sc}, and FF with $RbGeI_3$ absorber bandgap.

E_g (eV)	PCE (%)	V_{oc} (V)	J_{sc} (mA/cm^2)	FF (%)
1.3	19.75	0.803	30.79	79.89
1.4 ★	19.87	0.894	27.42	81.02
1.5	18.75	0.964	23.81	81.66
1.6	17.10	0.994	20.69	83.09
1.7	15.39	0.997	17.95	85.97

H. ETL Thickness

The TiO_2 ETL thickness influences electron extraction, carrier transport, and optical transmission. To investigate its effect, the TiO_2 thickness was varied from 0.01 μm to 0.09 μm while all other parameters were held constant. The results are shown in Fig. 10 and Table XII. The highest PCE of 21.12 % was obtained at an ETL thickness of 0.01 μm (10 nm), and the efficiency decreased as the ETL became thicker, reaching a local minimum of 18.72 % at 0.04 μm before partially recovering to 19.97 % at 0.09 μm. J_{sc} remained nearly constant at 30.46 mA/cm^2 throughout, indicating that the ETL thickness mainly affects V_{oc} and FF through the electron transport path rather than through optical absorption. A 10 nm TiO_2 layer was therefore adopted for the optimized device.

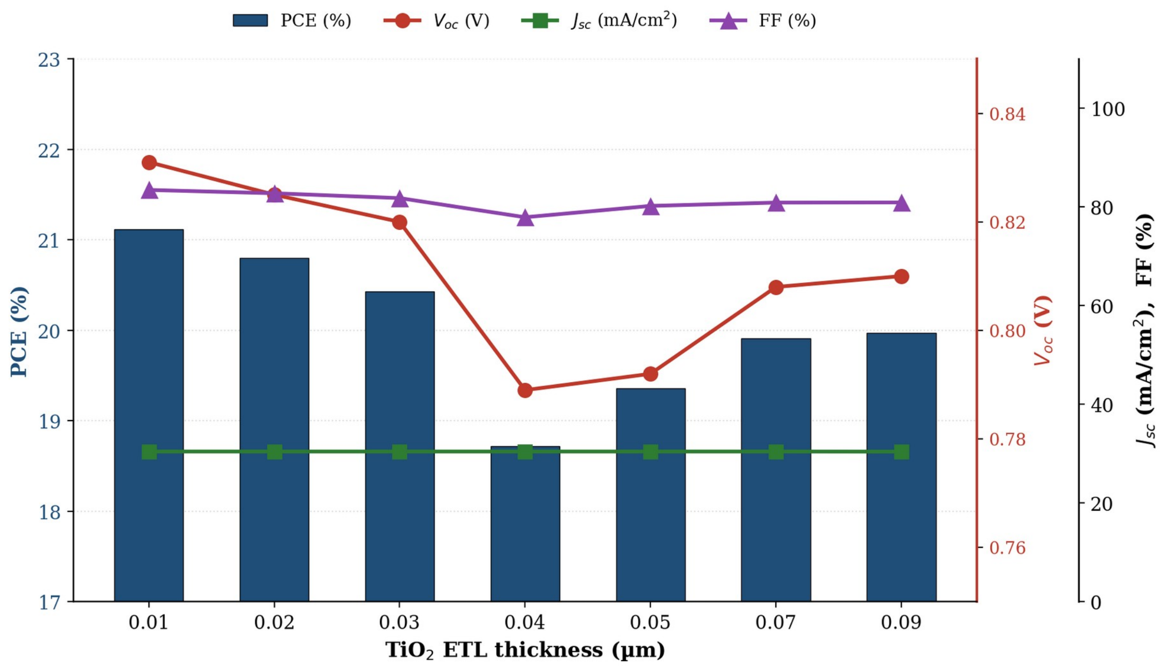

FIG. 10. Variation of PCE, V_{oc}, J_{sc}, and FF with the TiO_2 ETL thickness. The PCE is highest at the thinnest ETL (10 nm), reflecting the shorter electron transport path.

TABLE XII. Variation of PCE, V_{oc}, J_{sc}, and FF with TiO_2 ETL thickness.

Thickness (μm)	PCE (%)	V_{oc} (V)	J_{sc} (mA/cm^2)	FF (%)
0.01 ★	21.12	0.831	30.46	83.46
0.02	20.80	0.825	30.46	82.79
0.03	20.43	0.820	30.46	81.81
0.04	18.72	0.789	30.46	77.93
0.05	19.36	0.792	30.46	80.23
0.07	19.91	0.808	30.46	80.90
0.09	19.97	0.810	30.44	80.93

I. HTL Thickness

The CuI HTL thickness was swept from 0.1 μm to 1.9 μm to investigate its effect on device performance. The results are shown in Fig. 11 and Table XIII. In stark contrast to the ETL, the HTL thickness had almost no influence on device performance: PCE increased by less than 0.002 percentage points across the entire range, and the variations in V_{oc}, J_{sc}, and FF were similarly negligible. This is because CuI already provides effective hole transport at 0.1 μm thanks to its high hole mobility (~100 $cm^2V^{-1}s^{-1}$), and any further increase in thickness simply adds parasitic absorption without improving charge collection. A thickness of 100 nm was therefore selected for the final device to keep material usage low.

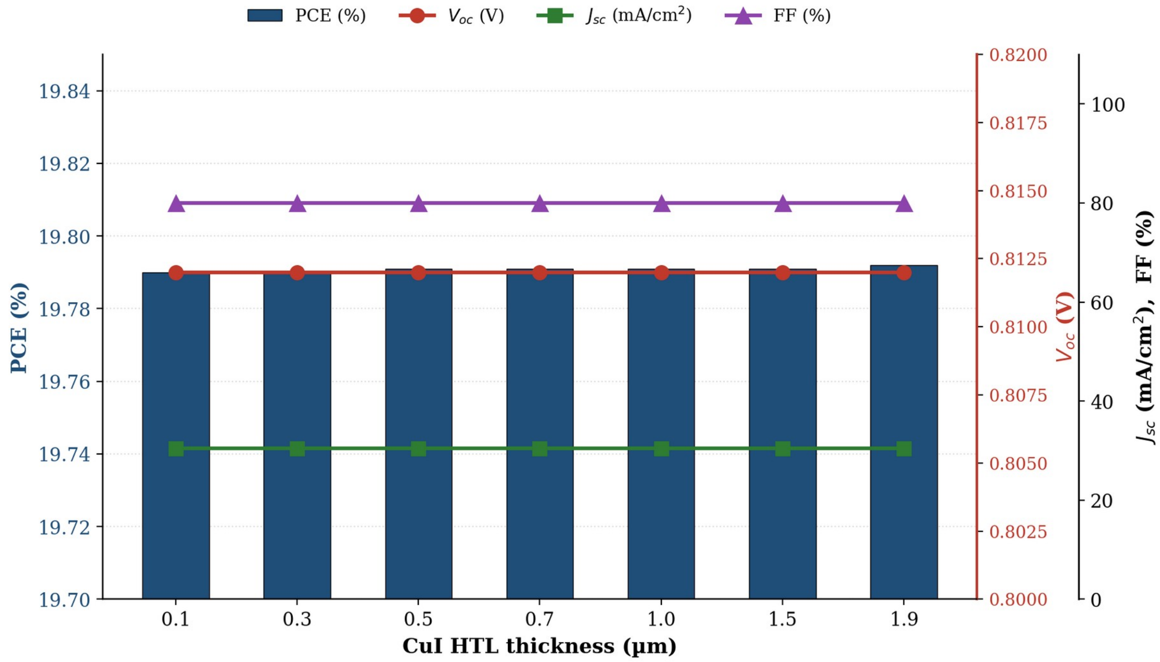

FIG. 11. Variation of PCE, V_{oc}, J_{sc}, and FF with the CuI HTL thickness. The HTL thickness has negligible influence on device performance.

TABLE XIII. Variation of PCE, V_{oc}, J_{sc}, and FF with CuI HTL thickness.

Thickness (μm)	PCE (%)	V_{oc} (V)	J_{sc} (mA/cm²)	FF (%)
0.1 ★	19.790	0.812	30.461	79.980
0.3	19.790	0.812	30.461	79.980
0.5	19.791	0.812	30.462	79.980
0.7	19.791	0.812	30.462	79.980
1.0	19.791	0.812	30.462	79.980
1.5	19.791	0.812	30.463	79.980
1.9	19.792	0.812	30.463	79.980

J. Conduction-Band Effective Density of States

The conduction-band effective density of states N_C influences the distribution of electrons in the conduction band and plays an important role in determining the electrical properties of the absorber layer. To investigate its effect, N_C was varied from 1×10^{17} to 1×10^{20} cm^{-3} while all other parameters were held constant. The results are shown in Fig. 12 and Table XIV. Increasing N_C reduced PCE gradually from 20.65 % at 1×10^{17} cm^{-3} to 18.15 % at 1×10^{20} cm^{-3}, with V_{oc} dropping from 0.859 V to 0.764 V. J_{sc} remained nearly unchanged at 30.46 mA/cm^2. The reduction in PCE is mainly associated with the decrease in V_{oc}: a higher N_C modifies the carrier concentration and shifts the position of the electron quasi-Fermi level, reducing the quasi-Fermi-level splitting. A value of N_C = 1×10^{17} cm^{-3} was therefore selected for the optimized device.

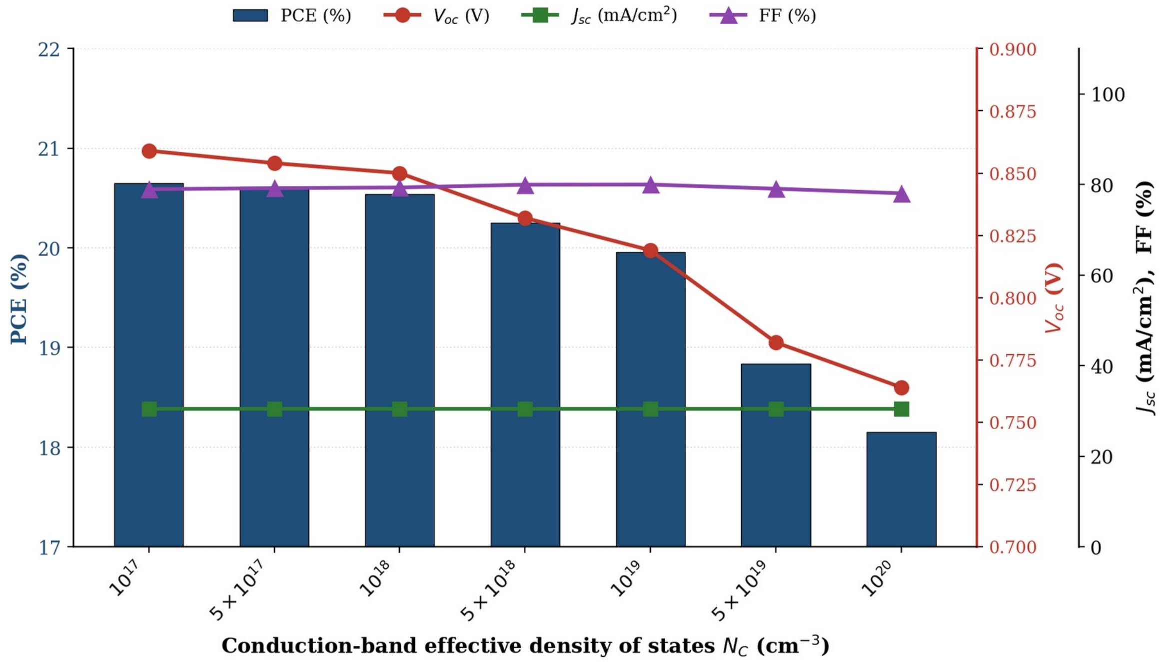

FIG. 12. Variation of PCE, V_{oc}, J_{sc}, and FF with the conduction-band effective density of states N_C.

TABLE XIV. Variation of PCE, V_{oc}, J_{sc}, and FF with conduction-band effective density of states N_C.

N_C (cm^{-3})	PCE (%)	V_{oc} (V)	J_{sc} (mA/cm^2)	FF (%)
1×10^{17} ★	20.65	0.859	30.46	78.94
5×10^{17}	20.61	0.854	30.46	79.18
1×10^{18}	20.54	0.850	30.46	79.32
5×10^{18}	20.25	0.832	30.46	79.94
1×10^{19}	19.96	0.819	30.46	79.97
5×10^{19}	18.84	0.782	30.46	79.05
1×10^{20}	18.15	0.764	30.46	78.04

K. Valence-Band Effective Density of States

The valence-band effective density of states N_V was similarly varied from 1×10^{17} to 1×10^{20} cm^{-3}. The results are shown in Fig. 13 and Table XV. The effect of N_V was more pronounced than that of N_C. PCE fell from 23.54 % at 1×10^{17} cm^{-3} to 18.86 % at 1×10^{20} cm^{-3}, with V_{oc} decreasing significantly from 0.944 V to 0.780 V. FF also decreased from 81.81 % to 79.40 %, while J_{sc} again remained nearly unchanged. These reductions arise because a higher N_V shifts the carrier distribution and reduces the quasi-Fermi-level splitting, in much the same way as observed for N_C but with a stronger effect on V_{oc}. A value of $N_V = 1\times10^{17}$ cm^{-3} was therefore selected for the optimized device.

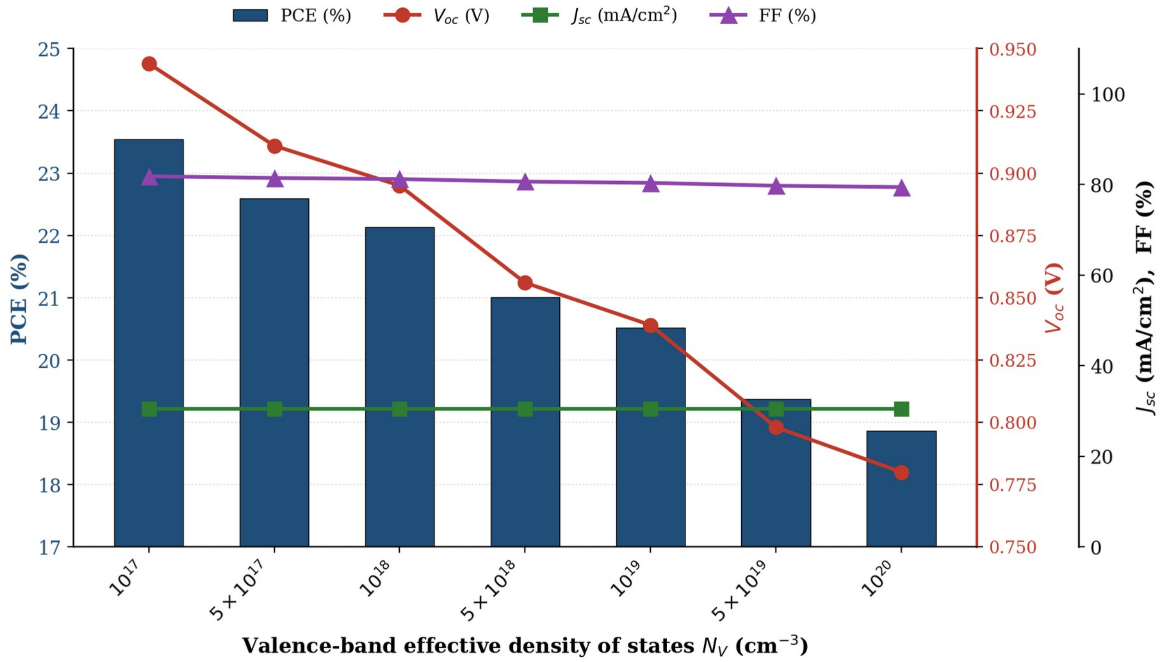

FIG. 13. Variation of PCE, V_{oc}, J_{sc}, and FF with the valence-band effective density of states N_V. The effect is more pronounced than for N_C.

TABLE XV. Variation of PCE, V_{oc}, J_{sc}, and FF with valence-band effective density of states N_V.

N_V (cm^{-3})	PCE (%)	V_{oc} (V)	J_{sc} (mA/cm^2)	FF (%)
1×10^{17} ★	23.54	0.944	30.46	81.81
5×10^{17}	22.59	0.911	30.46	81.42
1×10^{18}	22.13	0.895	30.46	81.18
5×10^{18}	21.01	0.856	30.46	80.62
1×10^{19}	20.52	0.839	30.46	80.34
5×10^{19}	19.37	0.798	30.46	79.71
1×10^{20}	18.86	0.780	30.46	79.40

L. Quantum Efficiency Analysis

The external quantum efficiency (EQE) spectrum of the optimized device is shown in Fig. 14. The EQE rises from about 34.5 % at 300 nm to 49.2 % at 350 nm and reaches approximately 100 % at 400 nm. The relatively lower EQE in the ultraviolet region can be attributed to surface recombination of high-energy photons and to optical losses in the FTO and TiO_2 front layers. Between 400 nm and 650 nm, the EQE remains on a broad plateau close to 99–100 %, indicating excellent photon absorption and efficient charge carrier collection within the visible region. As the wavelength increases beyond 650 nm, the EQE gradually decreases from 98.53 % at 650 nm to 76.17 % at 850 nm. A sharp drop in EQE is observed near 900 nm, where the EQE approaches zero. This behaviour corresponds to the absorption edge of the absorber material: photons with wavelengths greater than the cut-off wavelength ($\lambda_c \approx 1240/E_g \approx 886$ nm for $E_g = 1.4$ eV) do not have sufficient energy to excite electrons across the bandgap.

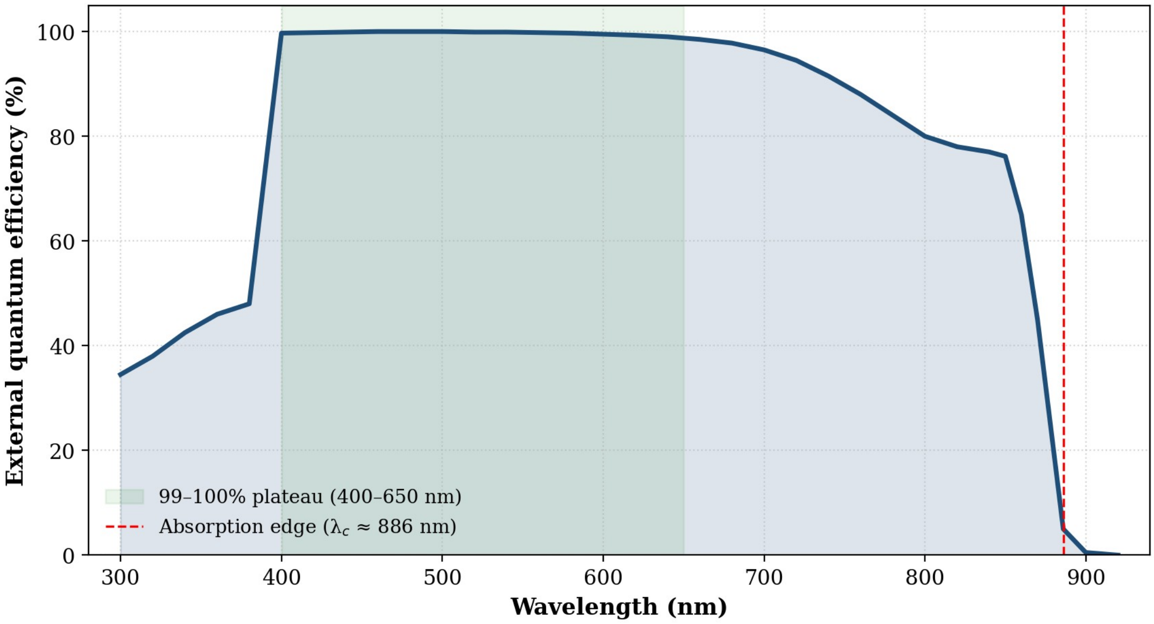

FIG. 14. Variation of external quantum efficiency (EQE) with incident wavelength for the optimized FTO/TiO_2/$RbGeI_3$/CuI/Au solar cell. The broad plateau close to 100 % between 400 and 650 nm confirms excellent visible-light response.

M. Energy Band Diagram Analysis

Fig. 15 shows the equilibrium energy band diagram of the optimized CuI/$RbGeI_3$/TiO_2/FTO device, with the horizontal axis representing the position across the device stack (total thickness 1310 nm: CuI 100 nm + $RbGeI_3$ 700 nm + TiO_2 10 nm + FTO 500 nm) and the vertical axis representing the electron energy (eV). At the CuI/$RbGeI_3$ interface, a pronounced conduction-band discontinuity is observed. Based on the electron affinity values of CuI (χ = 2.1 eV) and $RbGeI_3$ (χ = 3.9 eV), the conduction-band offset is approximately 1.8 eV, which acts as an effective electron-blocking barrier that suppresses electron back-transfer. The valence-band offset at the same interface is only about 0.1 eV, allowing holes to be extracted with minimal resistance.

Within the 700 nm $RbGeI_3$ absorber layer, both the conduction-band edge (E_c) and the valence-band edge (E_v) remain nearly constant, indicating a relatively uniform electrostatic potential throughout the bulk absorber. A second band discontinuity appears at the $RbGeI_3$/TiO_2 interface. Because the electron affinities of $RbGeI_3$ and TiO_2 are 3.9 eV and 4.0 eV respectively, the conduction-band offset is only about 0.10 eV, providing a small beneficial spike that suppresses electron back-transfer. The valence-band offset at this interface is approximately 1.99 eV, which blocks holes. The electron and hole quasi-Fermi levels remain well separated throughout the absorber, confirming the strong built-in field that drives charge separation and supports the high V_{oc} of 1.1127 V.

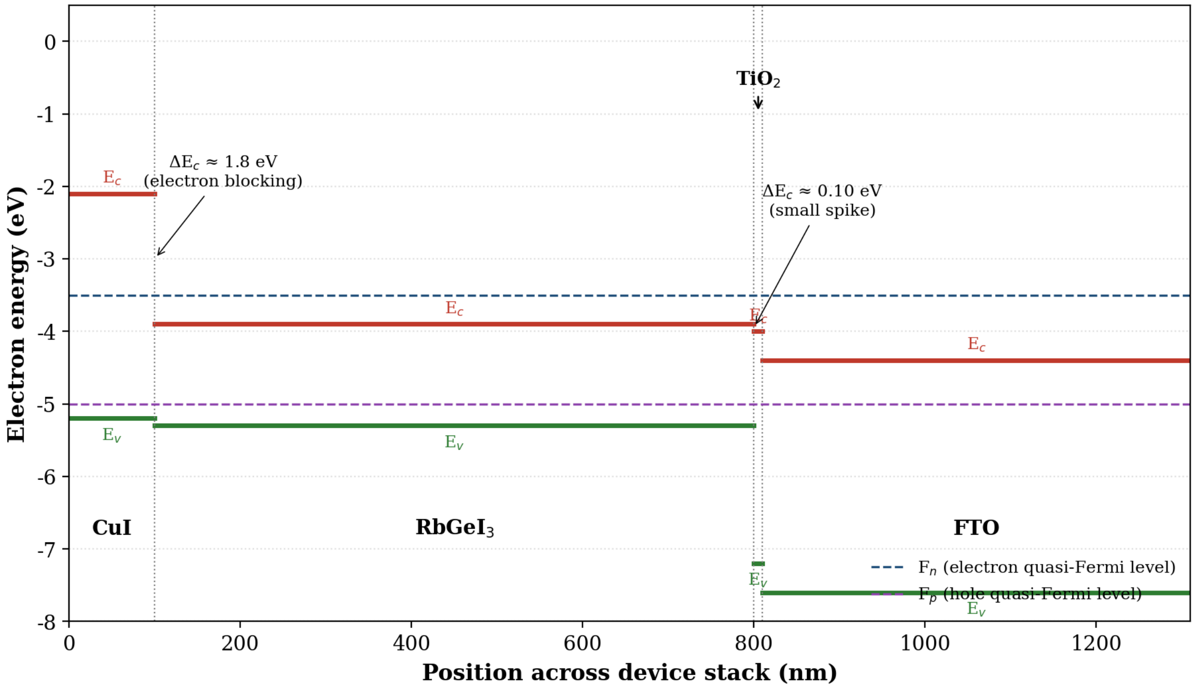

FIG. 15. Equilibrium energy band diagram of the optimized CuI/$RbGeI_3$/TiO_2/FTO solar cell, showing the conduction-band edge (E_c), valence-band edge (E_v), and quasi-Fermi levels (F_n, F_p) across the device stack.

N. Optimized Device Performance

After the sequential optimization of all eleven parameters, the final FTO/TiO_2/$RbGeI_3$/CuI/Au device achieves a PCE of 26.69 %, with V_{oc} = 1.1127 V, J_{sc} = 30.3156 mA/cm^2, and FF = 79.13 %. The optimized parameter set is summarized in Table XVI. The improvement from the initial 19.79 % of the unoptimized D4 device to the final 26.69 %, an absolute gain of 6.90 percentage points, is driven primarily by four factors: (i) the reduction of the absorber defect density to 10^{14} cm^{-3}, which suppresses SRH recombination; (ii) the choice of a 1.4 eV bandgap, which balances V_{oc} and J_{sc} near the Shockley–Queisser optimum; (iii) the optimization of the effective density of states to 1×10^{17} cm^{-3} for both N_C and N_V, which maximizes the quasi-Fermi-level splitting; and (iv) the adoption of a 10 nm TiO_2 ETL, which minimizes the electron transport path. The high V_{oc} and FF obtained here are consistent with previous simulation studies of CuI-based perovskite solar cells [33,35], which also reported favourable valence-band alignment and suppressed interfacial recombination at the CuI/absorber junction.

TABLE XVI. Optimized material and device parameters of the proposed FTO/TiO_2/$RbGeI_3$/CuI/Au solar cell.

Absorber thickness	700 nm	ETL (TiO_2) thickness	10 nm
Absorber bandgap	1.4 eV	HTL (CuI) thickness	100 nm
Absorber defect density	1×10^{14} cm^{-3}	Interface N_t (both)	1×10^{14} cm^{-2}
Electron affinity	3.9 eV	Final V_oc	1.1127 V
Dielectric constant	15	Final J_sc	30.32 mA/cm^2
N_C	1×10^{17} cm^{-3}	Final FF	79.13 %

N_V	1×10^{17} cm^{-3}	Final PCE	26.69 %

V. COMPARISON WITH LITERATURE

Table XVII compares the optimized FTO/TiO_2/$RbGeI_3$/CuI/Au device with recently reported $RbGeI_3$-based and related lead-free Ge/Sn-based perovskite solar cells. The proposed device delivers the highest PCE (26.69 %) and the highest V_{oc} (1.1127 V) among the simulation studies, exceeding the previous best $RbGeI_3$ result of 24.62 % reported by Loumachi et al. [28] by 2.07 percentage points and the 18.44 % reported by Raj et al. [29] by 8.25 percentage points. It also outperforms the experimental baseline of 10.11 % reported by Pindolia et al. [25] by more than 16 percentage points. The performance is competitive with the best tin-based lead-free devices (including the 26.40 % $CsSnI_3$ device of Park et al. [39] and the 25.86 % $MASnI_3$), while avoiding the $Sn^{2+} \rightarrow Sn^{4+}$ oxidation that limits the long-term stability of tin-based absorbers. The proposed PCE also exceeds that of CZTS-HTL lead-free devices reported by Piñón Reyes et al. [47], confirming the competitiveness of the CuI HTL approach adopted here.

TABLE XVII. Comparison of the proposed device with recently reported lead-free perovskite solar cells.

Device Structure	V_{oc} (V)	J_{sc} (mA/cm^2)	FF (%)	PCE (%)	Year [Ref.]
FTO/TiO_2/ $RbGeI_3$/NiO/Ag	0.531	28.89	63.68	10.11	2022 [25]
FTO/TiO_2/ $CsGeI_3$/CuI/Au	0.667	22.06	73.49	10.80	2023 [26]
ITO/C_6/ $RbGeI_3$/ CBTS/Au	0.99	33.20	82.80	24.62	2024 [28]
FTO/TiO_2/ $RbGeI_3$/ PEDOT:PSS/ Au	0.813	30.84	74.19	18.44	2025 [29]
FTO/TiO_2/ $MASnI_3$/ CBTS/Au	1.030	30.24	83.41	25.86	2026 [46]
FTO/TiO_2/ $CsSnI_3$/P3HT/ Au	0.972	34.70	78.21	26.40	2024 [39]
FTO/TiO_2/ $RbGeI_3$/CuI/Au (this work)	1.1127	30.32	79.13	26.69	This work

VI. LIMITATIONS AND FUTURE WORK

Although the simulated performance of the proposed device is encouraging, several limitations should be acknowledged. First, SCAPS-1D is a one-dimensional solver that assumes idealized conditions, including uniform material composition, abrupt and defect-free interfaces, and stable material parameters over the device lifetime. None of these assumptions is strictly guaranteed in experimental practice, where grain boundaries, interdiffusion, and degradation pathways can substantially degrade performance. Second, the defect densities adopted in this study (10^{14} cm^{-3} in the absorber and 10^{14} cm^{-2} at both interfaces), although realistic for well-passivated films, remain challenging to achieve consistently in $RbGeI_3$, where Ge^{2+} oxidation introduces deep defect states [30,31] during film preparation and operation. Third, the present study does not account for optical losses in the FTO and metal contact, for parasitic absorption in the transport layers, or for the temperature dependence of material parameters, all of which would reduce the achievable experimental PCE.

Translating the simulated 26.69 % PCE into an experimental device will require advances on several fronts. Ge^{2+} stabilization is the single most important obstacle to high-performance Ge-based perovskite solar cells, and effective encapsulation, controlled-atmosphere deposition, and the use of Ge^{2+}-stabilizing additives will be essential. The high sensitivity of the absorber/ETL interface to defect densities above 10^{14} cm^{-2} calls for dedicated passivation strategies (such as self-assembled monolayers (SAMs) [48,49], low-temperature atomic-layer-deposited interlayers, or chemical bath deposition of thin Al_2O_3) to suppress interfacial recombination. Anti-reflection coatings, surface texturing, plasmonic nanoparticles, and light-trapping structures could increase photon absorption without increasing absorber thickness. Future research should also investigate the influence of humidity, oxygen exposure, thermal stress, ultraviolet illumination, and ion migration on the performance degradation of $RbGeI_3$ devices, and the present one-dimensional electrical simulation could be complemented by coupled optical–electrical simulations [22,23] (e.g., using TCAD, COMSOL Multiphysics, or Lumerical) to capture the full carrier-transport, electric-field, and thermal behaviour. Finally, the optimized $RbGeI_3$ absorber, with its 1.4 eV bandgap, lies close to the Shockley-Queisser single-junction optimum and could also serve as a top cell in a tandem architecture with a wider-gap partner (e.g. a silicon or CIGS bottom cell) [3,45,50], although an ideal Si-tandem top cell would typically require a wider gap near 1.7 eV.

VII. CONCLUSION

This work has presented a comprehensive SCAPS-1D optimization of a lead-free $RbGeI_3$ absorber perovskite solar cell with the structure FTO/TiO_2/$RbGeI_3$/CuI/Au. Eight ETL–HTL configurations were screened by combining two electron transport layers (TiO_2 and PCBM) with four hole transport layers (NiO, CuI, CBTS, and Spiro-OMeTAD), and the TiO_2/$RbGeI_3$/CuI combination (device D4) was identified as the most promising. The eleven device parameters listed in Table XVI were then optimised sequentially under standard AM 1.5G illumination at 300 K, with the optimum value from each step carried forward as the input to the next.

The optimized device achieves a final PCE of 26.69 %, with V_{oc} = 1.1127 V, J_{sc} = 30.32 mA/cm^2, and FF = 79.13 %. The optimized device outperforms previously reported $RbGeI_3$ simulations and the experimental baseline, and remains competitive with the best tin-based lead-free devices while avoiding the Sn^{2+} oxidation problem. The quantum efficiency spectrum exhibits a broad plateau close to 100 % between 400 and 650 nm, and the equilibrium energy band diagram confirms favourable band alignment at both heterojunctions. The TiO_2/CuI transport-layer pair therefore emerges as a viable route to lead-free $RbGeI_3$ photovoltaics, and the parameter set listed in Table XVI can serve as a concrete starting point for experimental fabrication. Realising the simulated 26.69 % PCE in practice will depend on advances in Ge^{2+} stabilization, interface passivation, and encapsulation.

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